

HW3: Fairness & Bias Audit in Mental Health Data

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1. Setup

This project investigates fairness and bias in a mental health treatment dataset by comparing two machine learning models: Logistic Regression and a Neural Network (MLPClassifier). The dataset contains 341 rows and 17 columns. The goal is to predict treatment outcome and evaluate whether the models perform differently across protected groups. This is a binary classification task in which the model predicts whether a patient's treatment outcome improved or deteriorated.

The label Y used in this project is **Outcome**, which was encoded as:

- Improved = 1
- Deteriorated = 0

The protected attribute A is **Gender**, which was encoded as:

- Female = 0
- Male = 1

The feature set X excluded the protected attribute and included the following variables:

- Age
- Diagnosis
- Symptom Severity (1–10)
- Mood Score (1–10)
- Sleep Quality (1–10)
- Physical Activity (hrs/week)
- Medication

- Therapy Type
- Treatment Duration (weeks)
- Stress Level (1–10)
- Treatment Progress (1–10)
- AI-Detected Emotional State
- Adherence to Treatment (%)

Before training the models, categorical variables were converted using one-hot encoding. The dataset was then divided into training and testing sets using a 70/30 split with random state 42.

2. Results

The first part of the analysis compared the overall predictive performance of Logistic Regression and the Neural Network. Accuracy, precision, and recall were used as the main overall evaluation metrics.

Overall Model Performance

Model	Accuracy	Precision	Recall
Logistic Regression	0.5243	0.5625	0.4909
Neural Network (MLP)	0.4951	0.5254	0.5636

Table 1: Overall performance comparison between Logistic Regression and Neural Network

Table 1 shows the overall predictive performance of both models. Logistic Regression achieved a slightly higher accuracy of 0.5243, while the Neural Network had a slightly lower accuracy of 0.4951. However, the Neural Network had a higher recall of 0.5636 compared to 0.4909 for Logistic Regression, meaning it identified a larger proportion of actual positive cases. Overall, Logistic Regression performed slightly better in terms of accuracy and precision, while the Neural Network performed better in recall.

Group-Wise Fairness Metrics

To evaluate fairness, model predictions were separated by gender group. For each group, accuracy, true positive rate (TPR), and false positive rate (FPR) were calculated. TPR measures how well the model identifies actual positive cases, while FPR measures how often the model incorrectly predicts positive outcomes for negative cases.

Model	Group	Accuracy	TPR	FPR
Logistic Regression	Female	0.5400	0.5652	0.4815
Logistic Regression	Male	0.5094	0.4375	0.3810
Neural Network (MLP)	Female	0.4800	0.5652	0.5926
Neural Network (MLP)	Male	0.5094	0.5625	0.5714

Table 2: Fairness metrics by gender group

Table 2 shows that Logistic Regression performed somewhat differently across gender groups. For Logistic Regression, females had a higher TPR of 0.5652 compared to 0.4375 for males, meaning the model detected more true positive cases for females. Females also had a higher FPR of 0.4815 compared to 0.3810 for males, meaning the model also produced more false positives for females. This suggests that the male group was disadvantaged in terms of lower true positive rate.

For the Neural Network, the TPR values were much closer between the two groups, with 0.5652 for females and 0.5625 for males. The FPR values were also close, although both were fairly high, with 0.5926 for females and 0.5714 for males. This suggests that the Neural Network treated both groups more similarly, although its error rates were less desirable overall.

Model Fairness Comparison

Model	TPR Gap	FPR Gap
Logistic Regression	0.1277	0.1005
Neural Network (MLP)	0.0027	0.0212

Table 3: Fairness gap comparison across models

Table 3 summarizes the fairness gaps for each model. Logistic Regression had a TPR gap of 0.1277 and an FPR gap of 0.1005, showing a more noticeable difference between gender groups. In contrast, the Neural Network had a very small TPR gap of 0.0027 and a much

smaller FPR gap of 0.0212. Based on these fairness gaps, the Neural Network was fairer across gender groups, even though its overall performance was slightly worse.

3. Mitigation

To reduce bias, threshold adjustment was used as the fairness mitigation method. This approach changes the classification threshold separately for each protected group after the model has already been trained. Instead of using the same threshold for everyone, group-specific thresholds can be used to reduce disparities in TPR or FPR.

In this project, threshold adjustment was applied to the Logistic Regression model. The thresholds used were:

- Female threshold = 0.50
- Male threshold = 0.40

These values were chosen to reduce the fairness gap while keeping overall model performance reasonably stable.

Before vs After Mitigation by Group

Version	Group	Accuracy	TPR	FPR
Before Mitigation	Female	0.5400	0.5652	0.4815
Before Mitigation	Male	0.5094	0.4375	0.3810
After Mitigation	Female	0.5400	0.5652	0.4815
After Mitigation	Male	0.5283	0.6562	0.6667

Table 4: Group-wise fairness metrics before and after threshold adjustment

Table 4 shows that after threshold adjustment, the female group metrics stayed the same, while the male group changed noticeably. The male TPR increased from 0.4375 to 0.6562, which means the model became better at identifying true positive cases for males. However, the male FPR also increased sharply from 0.3810 to 0.6667, which means the model also produced many more false positives for males after mitigation.

Before vs After Fairness Gap Comparison

Version	TPR Gap	FPR Gap	Accuracy
Before Mitigation	0.1277	0.1005	0.5243
After Mitigation	0.0910	0.1852	0.5340

Table 5: Fairness gaps and accuracy before and after mitigation

Table 5 shows the trade-off created by threshold adjustment. After mitigation, the TPR gap decreased from 0.1277 to 0.0910, which indicates an improvement in equal opportunity fairness. At the same time, the FPR gap increased from 0.1005 to 0.1852, which means predictive equality became worse. Interestingly, the overall accuracy slightly increased from 0.5243 to 0.5340. This shows that the mitigation improved fairness with respect to TPR parity, but worsened fairness with respect to FPR parity.

4. Discussion

The results of this project show that model performance should not be judged only by overall accuracy. Even if a model performs reasonably well overall, it may still treat different groups unequally. This is why group-wise fairness metrics such as TPR and FPR are important.

Possible sources of bias in this dataset include imbalance in group representation, differences in feature distributions across gender groups, and indirect relationships between non-protected features and the protected attribute. Even though gender was excluded from the feature set, other variables may still capture related information and lead to unequal outcomes.

Comparing Logistic Regression and the Neural Network also shows that a more complex model does not always perform better overall, but it may produce more balanced outcomes across groups. In this case, the Neural Network had slightly lower overall performance but much smaller fairness gaps, making it the fairer model across gender groups.

The mitigation method improved fairness by reducing the difference in TPR between groups, but it also increased the difference in FPR. This demonstrates the trade-off between fairness and error types. A possible improvement for future work would be to test reweighting, tune thresholds more systematically, or examine fairness using another protected attribute such as age group.